

Learning Hub

ISIT307 – Assignment #1 Report

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Note: Bay Beng Siang's university email address was not yet available at the time this report was generated and must be filled in before submission.

Requirements, Remarks & Readme

Learning Hub is a kid-friendly quiz web application built with plain PHP (no framework), PHP sessions for tracking game state, and pipe-delimited (|) text files for persistent storage of quiz question pools and the leaderboard. No database is used, per the assignment brief.

How to run

Developed and tested using Sublime Text as the code editor and MAMP as the local PHP server.

Using MAMP:

1. Place (or symlink) the project folder inside MAMP's htdocs directory, e.g. /Applications/MAMP/htdocs/Assignment 1.
2. Open the MAMP app and click **Start Servers**.
3. Browse to `http://localhost:8080/Assignment 1/index.php` (adjust the port to whatever MAMP's Apache port is configured as, e.g. 8888 by default).

Using PHP's built-in server (alternative):

1. Ensure PHP 7.4+ is installed.
2. From the project's root folder, run: `php -S localhost:8000`
3. Open `http://localhost:8000/index.php` in a browser.

Game rules implemented

- The player enters a nickname at the start of a game; overall points start at 0.
- The player picks a topic (Math or Sea World) and can play unlimited quizzes, same or different topic, within one game.
- Each quiz randomly selects 3 questions (no repeats) from a pool of 8 questions per topic (minimum 6 required by the brief).

- Math questions show an addition or subtraction equation; the player types the numeric answer. An empty or wrong answer counts as incorrect.
- Sea World questions show a sea-animal image with a name below it; the player answers “Correct” or “Not Correct” as to whether the name matches the animal.
- Scoring per quiz: $\text{points} = (\text{correct} \times 3) - (\text{incorrect} \times 2)$.
- After each quiz: number correct, number incorrect, points from that quiz, and the running overall points for the current game are displayed.
- After a quiz the player can start a new quiz (same/different topic), view the Leaderboard, or Exit.
- The Leaderboard shows cumulative points per nickname across all games ever played, and can be sorted alphabetically by nickname or by highest score.
- On Exit, the current game's points are added to the player's cumulative leaderboard total (viewable via the Leaderboard); the screen itself shows the nickname and the points earned this game, plus a “Start New Game” option that clears the session and returns to the nickname screen.

Validation & edge cases

- Empty nickname is rejected client-side (HTML5 required) and server-side (PHP trim/empty check).
- Empty or non-numeric math answers are graded as incorrect rather than causing an error.
- All user-supplied output (nickname, leaderboard entries) is escaped with `htmlspecialchars()` to prevent XSS; nicknames are trimmed, stripped of HTML tags, capped at 30 characters, and have the "|" delimiter character removed so a nickname can never corrupt its own pipe-delimited leaderboard.txt record.
- leaderboard.txt is created/handled gracefully if missing or empty.

Website Files

The folder structure follows the brief's suggested layout:

File / Folder	Description
index.php	Nickname entry screen. Validates the nickname (client- and server-side), resets the session, and starts a new game with overall_points = 0.
menu.php	Main hub shown after the nickname is set. Links to the Math Quiz, Sea World Quiz, Leaderboard and Exit.
quiz.php	Loads the question pool for ?topic=math seaworld, picks 3 random non-repeating questions, stores them in the session, and renders the quiz form.
process_quiz.php	POST handler that grades the submitted answers against the session's stored questions, updates the game's overall points, and redirects to result.php.
result.php	Shows correct/incorrect counts, points from the quiz just played, and the running overall points for the current game, with links to play again, view the Leaderboard, or Exit.
leaderboard.php	Displays the cumulative all-time leaderboard, sortable by nickname (A-Z) or by score (highest first) via ?sort=name score.
exit.php	Adds the current game's points to the player's cumulative leaderboard total (once per game), then shows the nickname and the points earned this game.
new_game.php	Clears the PHP session entirely and redirects back to index.php so a new game can begin.
includes/functions.php	All helper functions: loading question pools, picking random questions, scoring, loading/saving/sorting the leaderboard, and input sanitisation/escaping.
includes/layout_top.php includes/layout_bottom.php	Shared HTML header and footer (navigation bar with nickname/points, page shell) included on every page.
css/style.css	Original light-themed "field journal" design: flat colour blocks, hand-drawn heading font, hard-offset card/button shadows, dot-grid paper background — no stock gradients or icon fonts.
data/math_questions.txt	Math question pool, one "equation answer" record per line (8 questions: addition and subtraction only, per the brief).
data/seaworld_questions.txt	Sea World question pool, one "image displayed_name correct_or_incorrect" record per line (10 questions).
data/leaderboard.txt	Persisted "nickname totalpoints" records — the cumulative Leaderboard. Ships empty; entries are added as games are played.
images/*.svg	Eight original, simple hand-drawn SVG sea-animal icons (dolphin, shark, turtle, octopus, jellyfish, whale, starfish, seahorse) — not photos, to avoid copyright issues.

Data file formats

data/math_questions.txt – equation|answer (+ or -)

```
7+6|13   9+9|18   18-9|9   6-14|-8   ...
```

data/seaworld_questions.txt – image|displayed_name|correct_or_incorrect

```
dolphin.svg|Dolphin|correct   shark.svg|Turtle|incorrect   ...
```

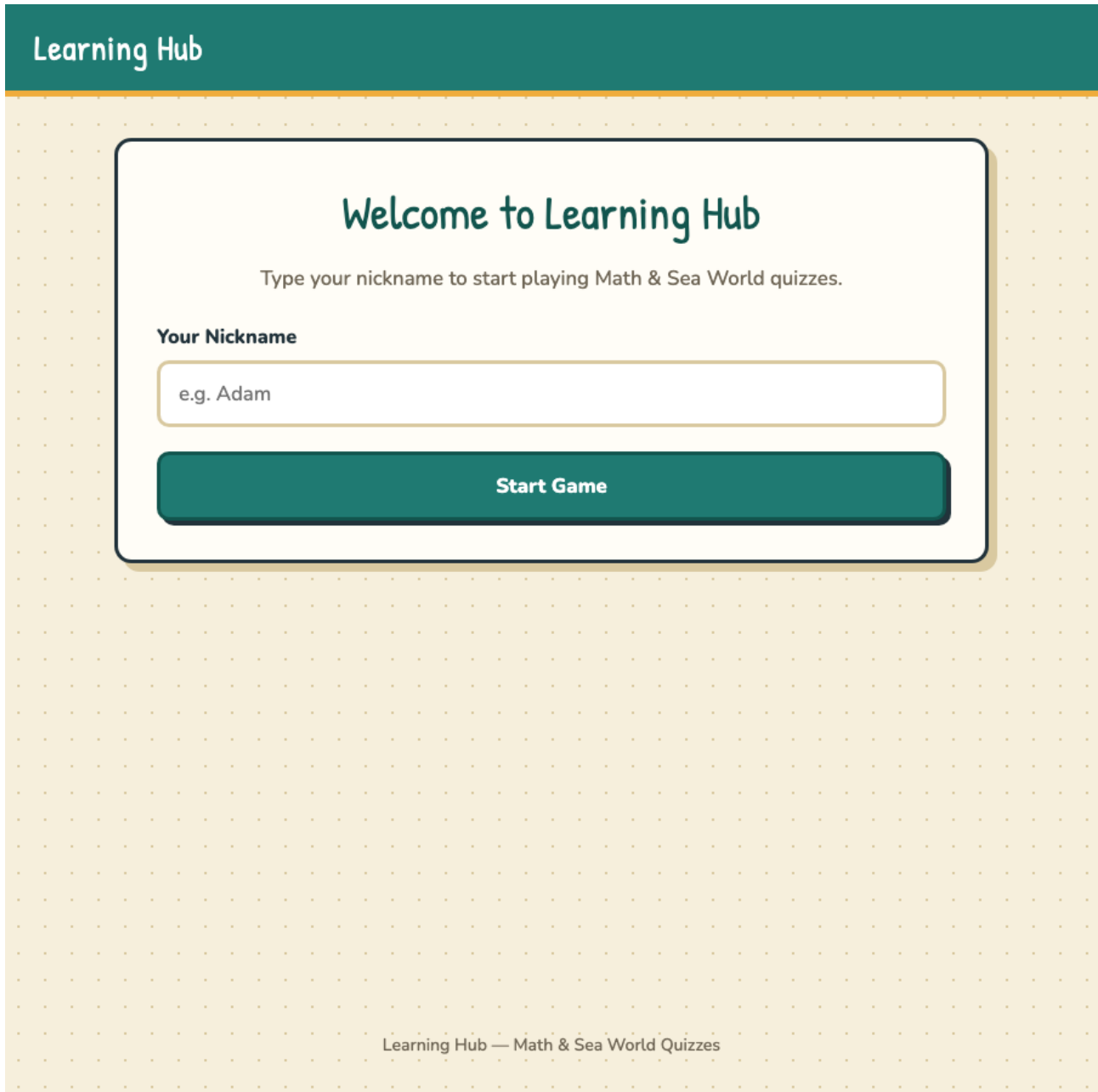
data/leaderboard.txt – nickname|totalpoints (starts empty)

```
Adam|18   (populated as players finish games)
```

User Manual

The screenshots below walk through a full play session: entering a nickname, the menu, playing a Math quiz and a Sea World quiz, viewing the Leaderboard both ways, and exiting the game.

1. Nickname Entry



The screenshot shows the 'Learning Hub' interface. At the top is a dark teal header with the text 'Learning Hub' in white. Below this is a light beige background with a subtle dot pattern. In the center is a white rounded rectangle with a dark teal border. Inside this rectangle, the text 'Welcome to Learning Hub' is displayed in a large, dark teal font. Below it, in a smaller dark teal font, is the instruction 'Type your nickname to start playing Math & Sea World quizzes.' Underneath this is the label 'Your Nickname' in bold dark teal. Below the label is a white text input field with a thin gold border, containing the placeholder text 'e.g. Adam'. Below the input field is a large, dark teal button with rounded corners and a dark teal border, containing the text 'Start Game' in white. At the bottom of the beige background, centered, is the text 'Learning Hub — Math & Sea World Quizzes' in a small, dark teal font.

The starting screen. The player types a nickname to begin a new game with 0 overall points.

2. Nickname Validation

Learning Hub

Welcome to Learning Hub

Type your nickname to start playing Math & Sea World quizzes.

Please type a nickname before starting!

Your Nickname

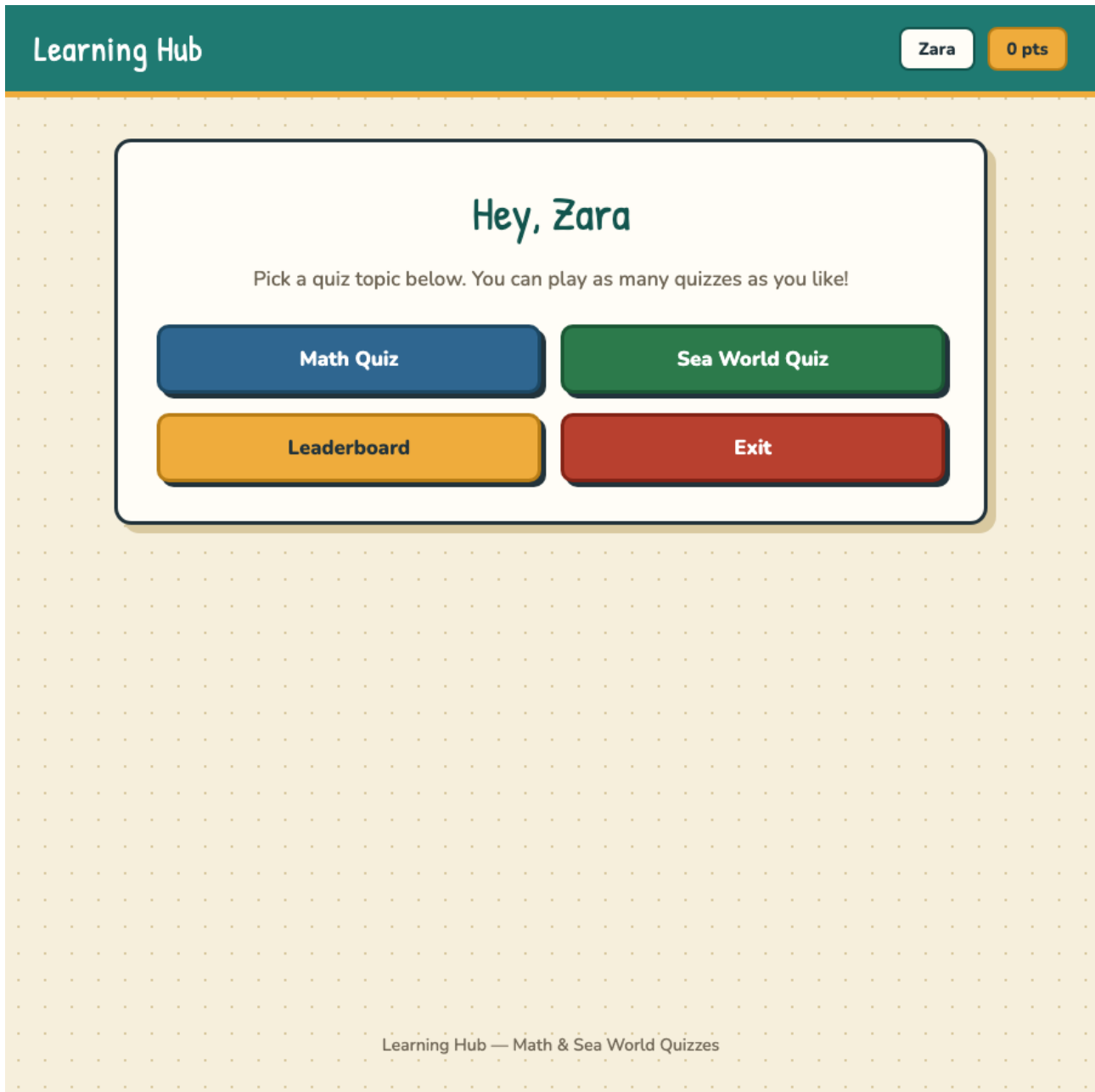
e.g. Adam

Start Game

Learning Hub — Math & Sea World Quizzes

Submitting an empty (or whitespace-only) nickname is rejected with an inline error message, both client-side (HTML5 required) and server-side.

3. Main Menu



After entering a nickname, the player lands on the menu with buttons for Math Quiz, Sea World Quiz, Leaderboard and Exit. The header always shows the current nickname and overall points for this game.

4. Math Quiz

Learning Hub

Zara0 pts

Math Quiz

Type the correct answer for each equation.

Question 1

$6 - 14 =$

Question 2

$25 - 8 =$

Question 3

$18 - 9 =$

Submit Answers

Learning Hub — Math & Sea World Quizzes

Three randomly chosen equations (addition or subtraction, per the brief) are displayed, each with a text box for the player's numeric answer.

5. Math Quiz Result

Learning Hub

Zara4 pts

Math Quiz Complete!

Nice work, Zara! Here's how you did:

2 Correct	1 Incorrect
4 Points This Quiz	4 Overall Points
Math Quiz	Sea World Quiz
Leaderboard	Exit

Learning Hub — Math & Sea World Quizzes

After submitting, the number correct/incorrect, points earned this quiz ($[correct] \times 3 - [incorrect] \times 2$), and the running overall points for the current game are shown, along with options to play again, view the Leaderboard, or Exit.

6. Sea World Quiz

Learning Hub

Zara4 pts

Sea World Quiz

Is the name below the animal correct or not correct?

Question 1



Whale

☐ Correct☐ Not Correct

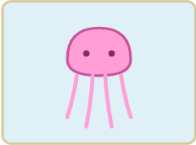
Question 2



Starfish

☐ Correct☐ Not Correct

Question 3



Jellyfish

☐ Correct☐ Not Correct

Submit Answers

Learning Hub — Math & Sea World Quizzes

Three randomly chosen sea-animal images are shown, each with a name below it. The player answers whether the displayed name is Correct or Not Correct for that animal.

7. Sea World Quiz Result

Learning Hub

Zara8 pts

Sea World Quiz Complete!

Nice work, Zara! Here's how you did:

2 Correct	1 Incorrect
4 Points This Quiz	8 Overall Points

Math Quiz

Sea World Quiz

Leaderboard

Exit

Learning Hub — Math & Sea World Quizzes

Same result layout as the Math quiz, with the overall points now accumulated across both quizzes played in this game.

8. Leaderboard – Sorted by Score

Learning Hub

Zara8 pts

Leaderboard

Total points earned across all games ever played.

Sort by Name (A-Z)

Sort by Score (High-Low)

#	Nickname	Points
1	Kevin	4
2	Adam	3
3	Sandy	-1

Return

Learning Hub — Math & Sea World Quizzes

The Leaderboard shows every nickname's cumulative points across all games ever played, sorted highest score first.

9. Leaderboard – Sorted by Name

Learning Hub

Zara8 pts

Leaderboard

Total points earned across all games ever played.

Sort by Name (A-Z)

Sort by Score (High-Low)

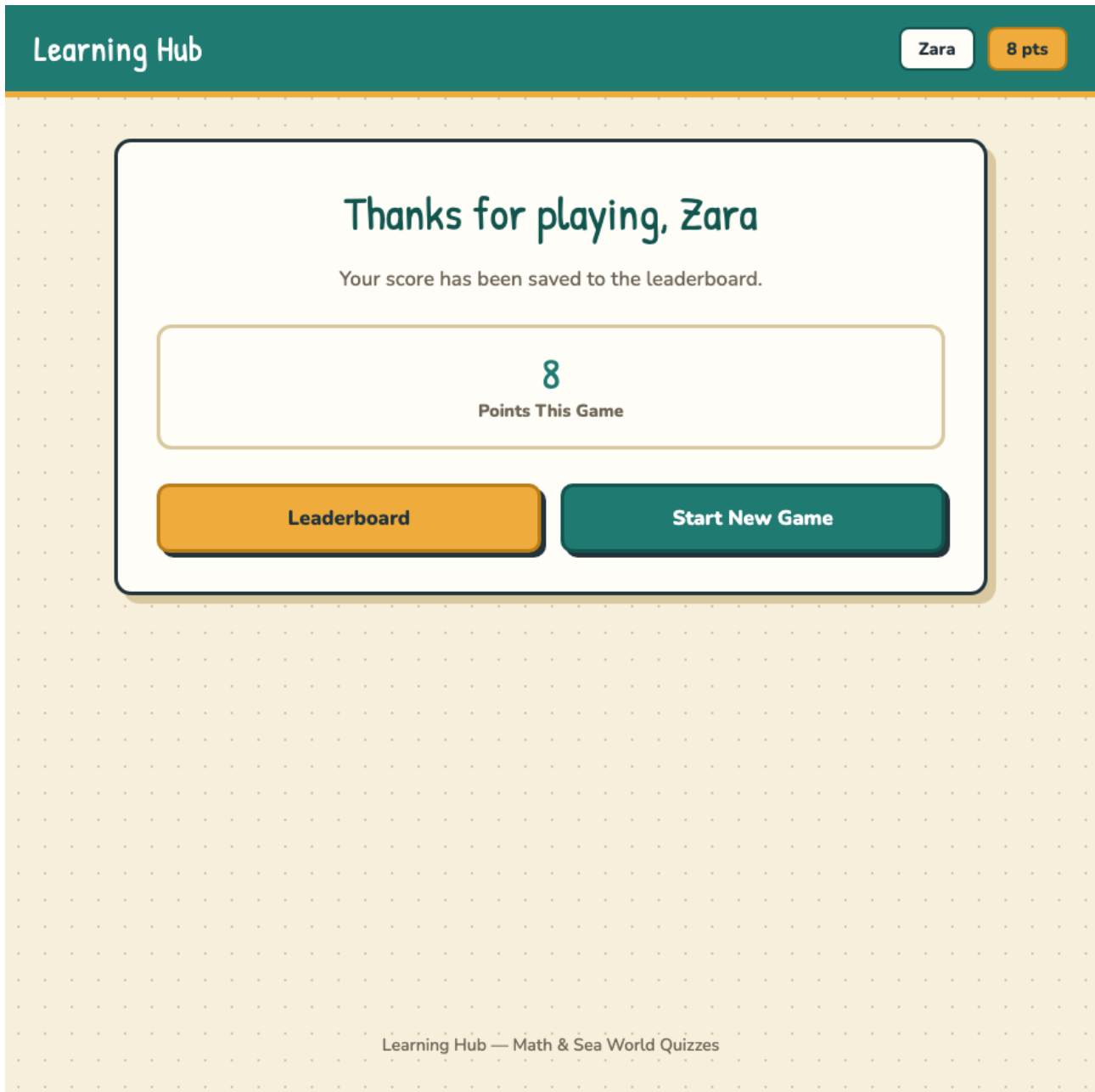
#	Nickname	Points
1	Adam	3
2	Kevin	4
3	Sandy	-1

Return

Learning Hub — Math & Sea World Quizzes

The same data re-ordered alphabetically by nickname (A–Z).

10. Exit Screen



Choosing Exit adds this game's points to the player's cumulative leaderboard total (saved to leaderboard.txt) and displays the nickname and the points earned this game, with a Start New Game option. The all-time cumulative total is viewable separately via the Leaderboard.

11. Start New Game

Learning Hub

Welcome to Learning Hub

Type your nickname to start playing Math & Sea World quizzes.

Your Nickname

e.g. Adam

Start Game

Learning Hub — Math & Sea World Quizzes

Clicking “Start New Game” clears the session and returns to the nickname entry screen, ready for the next player.